

5. Observational Studies

ISS5096 || ECI

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Roadmap

1. Identification in Observational Studies
2. Selection on Observables
3. Sensitivity Analysis
4. Partial Identification

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 - What if the assumptions are wrong? $\rightsquigarrow$ **sensitivity analysis** and **partial identification**.
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- Q: Why are observational studies in causal inference important? (What are the limitations of RCTs?)

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Source: Twitter @NobelPrize

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Scientific Background on the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021

ANSWERING CAUSAL QUESTIONS USING OBSERVATIONAL DATA

The Committee for the Prize in Economic Sciences in Memory of Alfred Nobel

*“Taken together, therefore, the Laureates’ contributions have played a central role in establishing the so-called design-based approach in economics. This approach – aimed at **emulating a randomized experiment to answer a causal question using observational data** – has transformed applied work and improved researchers’ ability to answer causal questions of great importance for economic and social policy using observational data.” (p.2)*

1/ Identification in Observational Studies

Randomized Experiment Review

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- **Randomized experiment** is an experiment with two properties:
 1. **Positivity:** assignment is probabilistic (and not deterministic):
 $0 < \mathbb{P}[D_i = 1] < 1$
 2. **Unconfoundedness:** $\mathbb{P}[D_i = 1 | \mathbf{Y}(1), \mathbf{Y}(0)] = \mathbb{P}[D_i = 1]$
 - Treatment assignment does not depend on any potential outcomes.
 - Sometimes written as $D_i \perp\!\!\!\perp (\mathbf{Y}(1), \mathbf{Y}(0))$.

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- Without unconfoundedness: naive diff-in-means = PATT + selection bias
- **Selection bias:** how different the treated and control groups are in terms of their potential outcome under control.

Selection Bias = Unidentified ATT

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 - Naive estimate: influencers do worse without comment sections than with them.
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- With an unbounded Y_i , we cannot even bound the ATT because, in principle, SB could be anywhere from $-\infty$ to ∞ .
- We say ATT (as well as ATE) are **unidentified** w/o further assumptions.

What is identification?

- **Identification** connects the counterfactual to the observed.
 - **Counterfactual distribution** $\mathbb{P}^*$ of $\{Y_i(1), Y_i(0), D_i, \mathbf{X}_i\}$.
 - **Observational distribution** $\mathbb{P}$ of $\{Y_i, D_i, \mathbf{X}_i\}$.
 - Causal quantities are functions of $\mathbb{P}^*$, but we get samples from $\mathbb{P}$.
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- Connecting counterfactuals to the observational requires **assumptions**.
 - **“What is your identification strategy?”** = what are the assumptions that allow you to claim that you’ve estimated a causal effect?
 - Research design can help justify assumptions (experiments, RDD, etc).
 - Or you will need to justify them through argument.

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- **Identification** comes first, then comes **estimation**.
 - Without identification, properties of the estimator are unimportant.
 - keep them separate: estimator shouldn't drive identification.

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 - Effect of corporate giants on economic development (confounder: previous economic development)
- Confounding $\rightsquigarrow$ incomplete identification of ATE $\rightsquigarrow$ biased estimators.
- What to do?

2/ Selection on Observables

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 - Also called: conditional unconfoundedness, weak ignorability, selection on observables, no omitted variables, exogenous, conditional exchangeability, etc.
 - $\rightsquigarrow$ Conditional on some covariates, D_i is (effectively) randomly assigned.
 2. **Positivity or Overlap:** $0 < \mathbb{P}[D_i = 1 | \mathbf{X}_i] < 1$
 - Treatment and control are both possible at every value of $\mathbf{X}_i$
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- These are assumptions that **can be wrong!**

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- How the mean of the potential outcomes vary with the covariates.
- Key part of the identification above:

$$\underbrace{\mu_1(\mathbf{x})}_{\text{counterfactual}} = \underbrace{\mathbb{E}[\textcolor{red}{Y}_i|D_i = 1, \mathbf{x}_i = \mathbf{x}]}_{\text{observational}}, \quad \mu_0(\mathbf{x}) = \mathbb{E}[\textcolor{red}{Y}_i|D_i = 0, \mathbf{x}_i = \mathbf{x}]$$

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- Procedure:
 1. Obtain predicted values for all units when $D_i = 1$.
 2. Obtain predicted values for all units when $D_i = 0$.
 3. Take the average difference between these predicted values.

Coefficients?

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- These make two very different assumptions about the CEFs!

Variance Estimation

- How do we get estimates of the variance of $\widehat{\tau}_{\text{reg}}$?

Variance Estimation

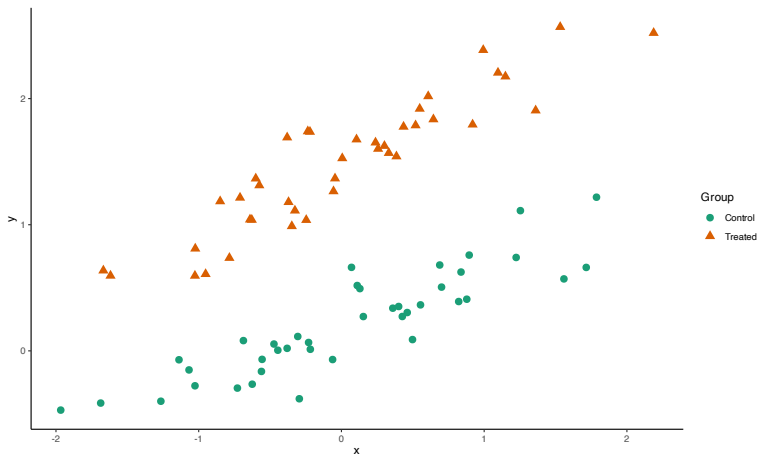
- How do we get estimates of the variance of $\widehat{\tau}_{\text{reg}}$?
- If an OLS coefficient $\rightsquigarrow$ use EHW variance estimator.
- Analytic expressions can be derived, but complicated!

Variance Estimation

- How do we get estimates of the variance of $\hat{\tau}_{\text{reg}}$?
- If an OLS coefficient $\rightsquigarrow$ use EHW variance estimator.
- Analytic expressions can be derived, but complicated!
- Computational alternative: **(nonparametric) bootstrap**
 - Randomly resample n rows of the data with replacement.
 - Refit the regressions on the bootstrapped data.
 - Calculate $\hat{\tau}_{\text{reg}}$ in each bootstrap.
 - Repeat several times and use empirical variance of the bootstraps.

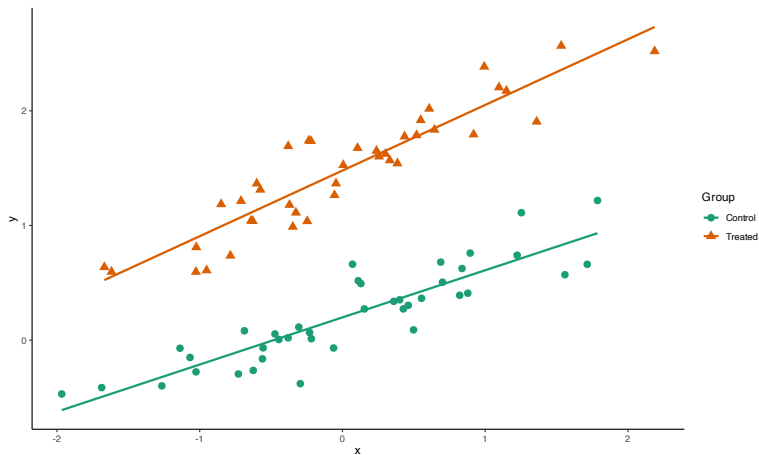
Imputation Estimator Visualization

```
1 > toy_data <- read_csv("https://bit.ly/3v0y2Ao")
```



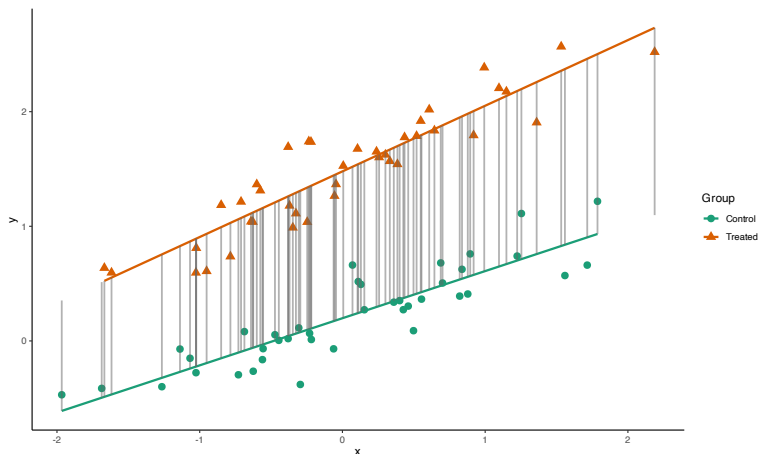
Imputation Estimator Visualization

```
1 > lm0 <- lm(y~x, data = toy_data, subset = d==0); lm1 <- lm(y~x, data = toy_data, subset = d==1)
```



Imputation Estimator Visualization

```
1 > mu0.imps = predict(lm0, toy_data); mu1.imps = predict(lm1, toy_data)
2 > cat("Estimate of ATE:", mean(mu1.imps - mu0.imps))
3 Estimate of ATE: 1.285176
```



Fully Interacted OLS & Imputation Estimator

- What if $\widehat{\mu}_1(\mathbf{x})$ and $\widehat{\mu}_0(\mathbf{x})$ are from fully interacted OLS with centered covariates?
 - Equivalent to running separate models for $\widehat{\mu}_1(\mathbf{x})$ and $\widehat{\mu}_0(\mathbf{x})$ (i.e., imputation estimator).
 - Why? The model $Y_i = \alpha + \tau D_i + \beta' \widetilde{\mathbf{X}}_i + \gamma'(D_i \widetilde{\mathbf{X}}_i) + \varepsilon_i$ gives:

$$D = 1 : \mathbb{E}[Y|\widetilde{\mathbf{X}}] = (\alpha + \tau) + (\beta + \gamma)' \widetilde{\mathbf{X}} \quad D = 0 : \mathbb{E}[Y|\widetilde{\mathbf{X}}] = \alpha + \beta' \widetilde{\mathbf{X}}$$

```
1 > toy_data$x_tilde <- toy_data$x - mean(toy_data$x)
2 > mod_full <- lm(y ~ d + x_tilde + d * x_tilde, data = toy_data)
3
4 > cat("\nEstimate of ATE (Imputation):", mean(mu1.imps - mu0.imps),
5       "\nEstimated coefficient on Di from full int.", mod_full$coefficients["d"])
6
7 Estimate of ATE (Imputation): 1.285176
8 Estimated coefficient on Di from full int. 1.285176
```

- $\rightsquigarrow$ With centered covariates, $\widehat{\tau}_{\text{reg}} \equiv$ estimated coefficient on D_i .
 - Would be the same for uninteracted model, except the variance will be larger (less precision).

Variance Estimation w/ Bootstrap

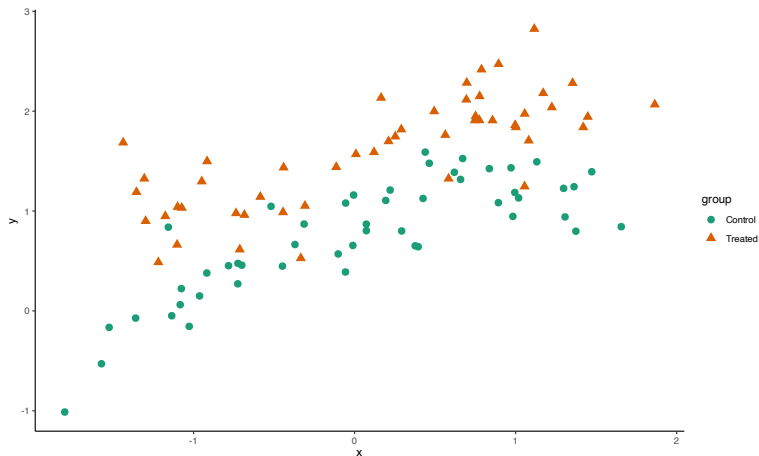
- Again, how do we get estimates of the variance of $\hat{\tau}_{\text{reg}}$?
 - (Nonparametric) bootstrap: a simulation-based alternative to analytic standard errors.
 - Idea: resample from the data to approximate the distribution of $\hat{\tau}_{\text{reg}}$.

```
1 > set.seed(02138); sims<-500; tau_hat_draws<-rep(NA, sims)
2 > for (i in 1:sims) { # Repeat the following several times
3   # 1. Randomly resample n rows of the data with replacement
4   sample_boot <- dplyr::slice_sample(toy_data, n = nrow(toy_data), replace = TRUE)
5
6   # 2. Refit the regressions on the bootstrapped data
7   model <- lm(y ~ d + x_tilde + d*x_tilde, data = toy_data)
8   dat1 <- sample_boot; dat1$d <- 1
9   dat0 <- sample_boot; dat0$d <- 0
10  mu1_hat <- predict(model, newdata = dat1)
11  mu0_hat <- predict(model, newdata = dat0)
12
13  # 3. Calculate tau_hat in each bootstrap
14  tau_hat_draws[i] <- mean(mu1_hat - mu0_hat)
15 }
16
17 > # 4. Use empirical variance of the bootstraps
18 > var(tau_hat_draws)
19 [1] 0.000254049
```

Nonlinear Relationships

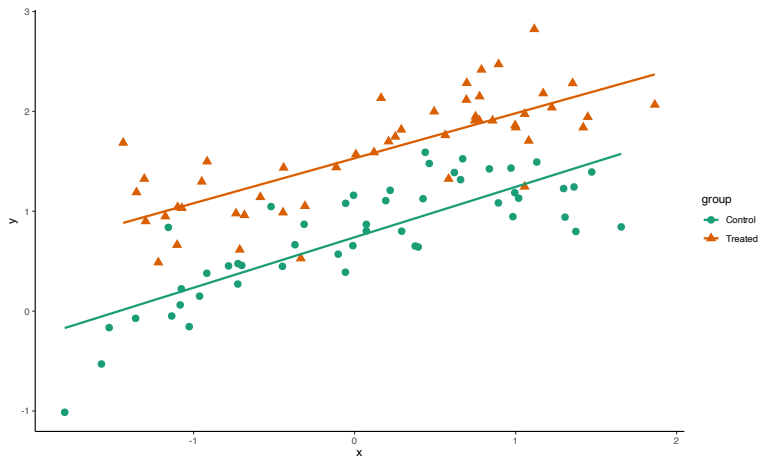
- Same idea but with nonlinear relationship between Y_i and X_i :

```
1 > toy_data_02 <- read_csv("https://bit.ly/4c0g0Pl") # sim data with nonlinear relationship
```



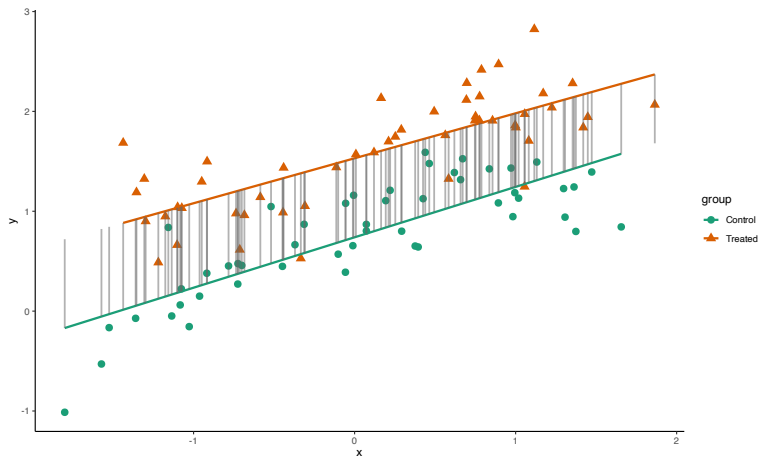
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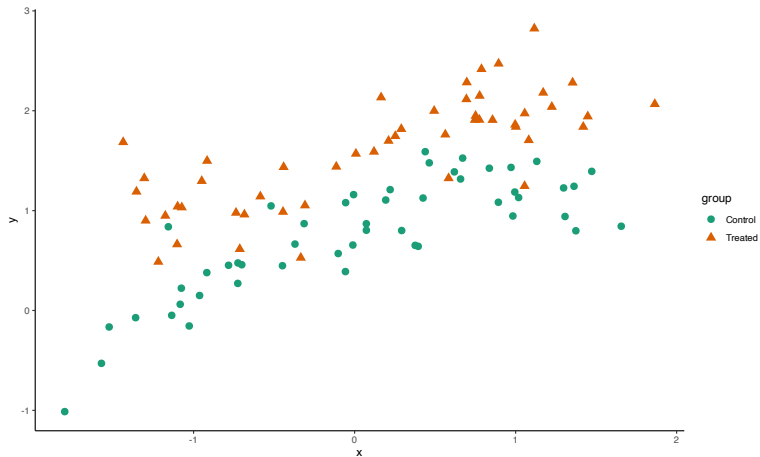


Using Semiparametric Regression

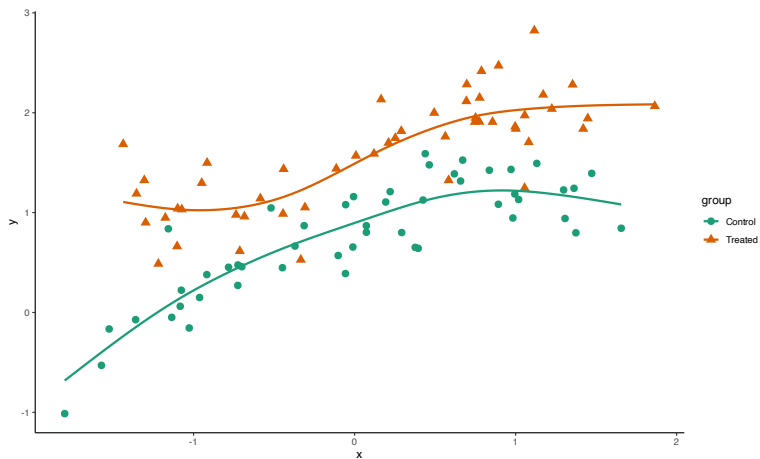
- Here, CEFs are nonlinear, but we don't know their form.
- We can use generalized additive models (GAMs) from the mgcv package for 'flexible' estimation:

```
1 > library(mgcv)
2 > gam1 <- gam(y~s(x), data = toy_data_02, subset = group=="Treated")
3 > gam0 <- gam(y~s(x), data = toy_data_02, subset = d==0); summary(gam0)
4
5 Family: gaussian
6 Link function: identity
7
8 Formula:
9 y ~ s(x)
10
11 Parametric coefficients:
12             Estimate Std. Error t value Pr(>|t|)
13 (Intercept)   0.2167      0.0307   7.059 2.05e-08 ***
14 ---
15 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
16
17 Approximate significance of smooth terms:
18             edf Ref.df    F p-value
19 s(x)         1      1 140.9 <2e-16 ***
20 ---
21 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
22
23 R-sq.(adj) = 0.782   Deviance explained = 78.8%
24 GCV = 0.03969   Scale est. = 0.037705   n = 40
```

Using GAMS



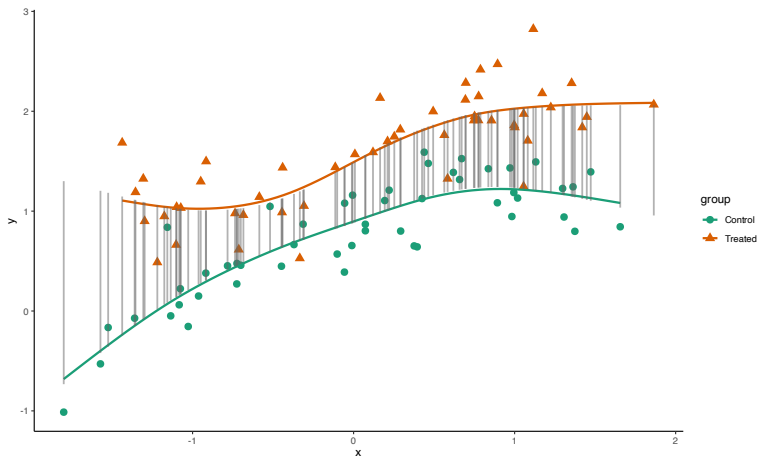
Using GAMS



Using GAMS

- We can estimate $\hat{\tau}_{\text{reg}}$ using the imputation estimator.

```
1 > cat("Estimate of ATE (GAM):",mean(predict(gam1) - predict(gam0)))  
2  
3 Estimate of ATE (GAM): 0.8379884
```



3/ Sensitivity Analysis

Cf. Cinelli & Hazlett (2020), *Making Sense of Sensitivity*, JRSS-B

Sensitivity Analysis for Regression

- Standard regression estimator of the ATE:

$$Y_i = \hat{\alpha} + \hat{\tau}D_i + \mathbf{X}'_i\hat{\beta} + \varepsilon_i$$

- Note: applies equally to fully interacted models (cf. W4).

Sensitivity Analysis for Regression

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$$Y_i = \alpha + \tau D_i + \mathbf{X}'_i\beta + \gamma U_i + \varepsilon_i$$

- Standard **omitted variable bias (OVB)** formula:

$$\begin{aligned}\hat{\tau} &= \tau + \gamma \times \underbrace{\frac{\text{cov}(D_i^{\perp \mathbf{X}}, U_i^{\perp \mathbf{X}})}{\mathbb{V}(D_i^{\perp \mathbf{X}})}}_{\text{regression of } U_i^{\perp \mathbf{X}} \text{ on } D_i^{\perp \mathbf{X}}} \\ &= \tau + \underbrace{\gamma \times \delta}_{\text{OVB}}\end{aligned}$$

- Gives us $\rightsquigarrow \tau = \hat{\tau} - \gamma\delta$

Partial R-Squared Interpretations

- Standard OVB written in terms of regression coefficients is difficult to reason about.
- Easier to reason with partial R^2 version of OVB (Cinelli and Hazlett, JRSSB 2020).
 - Intuition: bias $\propto$ how much U predicts $Y \times$ how much U predicts D .

$$|\text{bias}| = \sqrt{\frac{R^2_{Y \sim U | D, \mathbf{x}} \cdot R^2_{D \sim U | \mathbf{x}}}{1 - R^2_{D \sim U | \mathbf{x}}}} \cdot \frac{\mathbb{V}(Y \perp\!\!\!\perp \mathbf{x}, D)}{\mathbb{V}(D \perp\!\!\!\perp \mathbf{x})}$$

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- Partial R^2 is the incremental predictive value of one variable:

$$R_{Y \sim U|D, \mathbf{x}}^2 = \frac{R_{Y \sim D + \mathbf{x} + U}^2 - R_{Y \sim D + \mathbf{x}}^2}{1 - R_{Y \sim D + \mathbf{x}}^2} = \frac{\text{additional variance explained by } U}{\text{variance unexplained by } D, \mathbf{x}}$$

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- **Sensitivity analysis** can then vary two unknown parameters:
 - $R^2_{Y \sim U|D, \mathbf{x}} \in [0, 1]$: incremental predictive value of U for the outcome.
 - $R^2_{D \sim U|\mathbf{x}} \in [0, 1]$: incremental predictive value of U for treatment.
 - From these we can determine the bias and thus the true value of τ .

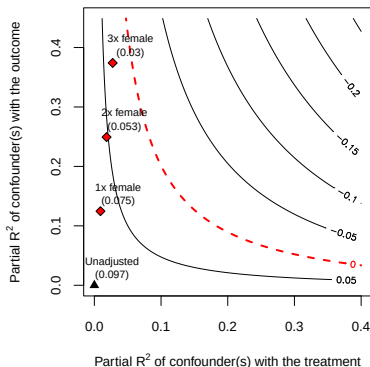
R Package: sensemakr

```
1 # load package
2 library(sensemakr)
3
4 # import dataset
5 data("darfur")
6
7 # run the regression model
8 model <- lm(peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
9             pastvoted + hhsize_darfur + female + village, data = darfur)
10
11 # conduct the sensitivity analysis
12 sensitivity <- sensemakr(model = model,
13                           treatment = "directlyharmed",
14                           benchmark_covariates = "female",
15                           kd = 1:3)
```

- Darfur conflict (Sudan, 2003) survey: D = directly harmed by violence, Y = support for peacemaking.
- female is the strongest known confounder $\rightsquigarrow$ used as benchmark.
 - kd (knock-on distance): how many times stronger would the unobserved confounder need to be (relative to female) to change our conclusion?

Sensitivity Analysis with Partial R^2

```
1 # plot bias contour of point estimate  
2 plot(sensitivity)
```



- Contour lines = adjusted estimate $\hat{\tau}$.
- Red diamonds = benchmarks at 1–3 $\times$ strength of female.

- Here: even a confounder 3 $\times$ as strong as female would not eliminate the effect. $\rightsquigarrow$ Our estimate is robust to plausible levels of unobserved confounding.

4/ Partial Identification

Cf. Manski (2003), *Partial Identification of Probability Distributions*

No-Assumption Bounds

- **Law of decreasing credibility** (Manski): credibility of inferences decreases with strength of assumptions
 - Idea: pick assumptions and then figure out what you can learn.
 - May not be point identified, but maybe we can **bound** the effect.

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- Can we improve using data? Rewrite the ATE with $p = \mathbb{P}(D_i = 1)$:

$$\begin{aligned}\tau &= \mathbb{E}[Y_i \mid D_i = 1]p + \mathbb{E}[Y_i(1) \mid D_i = 0](1 - p) \\ &\quad - \mathbb{E}[Y_i(0) \mid D_i = 1]p - \mathbb{E}[Y_i \mid D_i = 0](1 - p)\end{aligned}$$

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- Plug in y_L and y_U for the counterfactual means to get bounds for τ :

$$\tau \geq \mathbb{E}[Y_i | D_i = 1]p + y_L(1 - p) - y_U p - \mathbb{E}[Y_i | D_i = 0](1 - p)$$

$$\tau \leq \mathbb{E}[Y_i | D_i = 1]p + y_U(1 - p) - y_L p - \mathbb{E}[Y_i | D_i = 0](1 - p)$$

- These bounds have width of $|y_U - y_L|$, i.e., half of the logical bounds.
- But always will contain 0. Weak assumptions $\rightsquigarrow$ weak inferences.

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- **MTR** (Monotone Treatment Response): treatment doesn't hurt anyone.
 - $Y_i(1) \geq Y_i(0)$ for all $i \rightsquigarrow \tau \geq 0$.
 - E.g., job training won't make you *worse* at finding a job.
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 - People with higher baseline outcomes are more likely to seek treatment.
 - Formalizes the **direction** of selection bias as an assumption.
- Each assumption contributes a *different* piece of information:
 - MTR $\rightsquigarrow$ **lower bound**: the effect is not negative.
 - MTS $\rightsquigarrow$ **upper bound**: the effect is no larger than the observed gap.

How Does MTS Tighten the Bound?

- Each potential outcome mean has an **observed** and **unobserved** part:

$$\mathbb{E}[Y(1)] = \underbrace{\mathbb{E}[Y_i|D_i = 1]}_{\text{observed}} \cdot p + \underbrace{\mathbb{E}[Y_i(1)|D_i = 0]}_{\text{unobserved}} \cdot (1 - p)$$

$$\mathbb{E}[Y(0)] = \underbrace{\mathbb{E}[Y_i(0)|D_i = 1]}_{\text{unobserved}} \cdot p + \underbrace{\mathbb{E}[Y_i|D_i = 0]}_{\text{observed}} \cdot (1 - p)$$

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- MTS replaces **logical bounds** with tighter **data-driven bounds**:

To find the upper bound...	No assumptions	With MTS
Largest $\mathbb{E}[Y(1) D = 0]$ possible	$\leq y_U$	$\leq \mathbb{E}[Y_i D_i = 1]$
Smallest $\mathbb{E}[Y(0) D = 1]$ possible	$\geq y_L$	$\geq \mathbb{E}[Y_i D_i = 0]$

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- For the **lower bound**, directions reverse (smallest $\mathbb{E}[Y(1)]$, largest $\mathbb{E}[Y(0)]$) $\rightsquigarrow$ MTS constraints don't bind; MTR provides the lower bound instead.

Numerical Example: How Bounds Narrow

Setup: $Y \in [0, 1]$, $p = \mathbb{P}(D_i = 1) = 0.4$, $\mathbb{E}[Y_i | D_i = 1] = 0.7$,
 $\mathbb{E}[Y_i | D_i = 0] = 0.5$.

Assumptions	Bounds for τ	Width	What changed?
No assumptions (data only)	$[-0.42, 0.58]$	1.00	
+ MTR	$[0, 0.58]$	0.58	lower bound $\uparrow$
+ MTR + MTS	$[0, 0.20]$	0.20	upper bound $\downarrow$



Confidence Regions for Bounds

- More general setup:
 - True bounds $[\delta_L, \delta_U]$: the **identification region**.
 - Estimated bounds $[\hat{\delta}_L, \hat{\delta}_U]$ with standard errors $\widehat{\text{se}}(\hat{\delta}_L), \widehat{\text{se}}(\hat{\delta}_U)$.

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- Confidence interval:

$$\left[\hat{\delta}_L - z_{1-\alpha} \widehat{\text{se}}(\hat{\delta}_L), \quad \hat{\delta}_U + z_{1-\alpha} \widehat{\text{se}}(\hat{\delta}_U) \right]$$

- If $\tau = \delta_L$ or $\tau = \delta_U$ (boundary): coverage $\rightarrow 1 - \alpha$.
- If $\delta_L < \tau < \delta_U$ (interior): coverage $\rightarrow 1$.
- Uses one-sided $z_{1-\alpha}$ (not $z_{1-\alpha/2}$) because each bound is one-sided.

Summing Up

- Conditional unconfoundedness is an assumption on unmeasured data, and thus inherently untestable!
 - The data are uninformative about the distribution of $Y(0)$ for treated units and $Y(1)$ for control units.
 - The assumption, however, can be 'indirectly' assessed. $\rightsquigarrow$ sensitivity analysis

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 - Credibility–informativeness tradeoff (Manski).
- Up next: DAGs, i.e., a graphical framework for reasoning about which covariates to condition on.

Onto the presentations & discussions!

Contact Information:

jaewon.yoo@iss.nthu.edu.tw

<https://j1yoo.github.io/>



Confidence Regions for the Identification Region

- An alternative goal: cover the **identified region** $[\delta_L, \delta_U]$ itself with probability $1 - \alpha$:

$$\mathbb{P}(\widehat{\delta}_L \leq \delta_L, \widehat{\delta}_U \geq \delta_U) \geq 1 - \alpha$$

- Confidence interval:

$$\left[\widehat{\delta}_L - z_{1-\alpha/2} \widehat{\text{se}}(\widehat{\delta}_L), \widehat{\delta}_U + z_{1-\alpha/2} \widehat{\text{se}}(\widehat{\delta}_U) \right]$$

- Works because of the **Bonferroni inequality**:

$$\mathbb{P}(\widehat{\delta}_L \leq \delta_L \text{ and } \widehat{\delta}_U \geq \delta_U) \geq \mathbb{P}(\widehat{\delta}_L \leq \delta_L) + \mathbb{P}(\widehat{\delta}_U \geq \delta_U) - 1 = 1 - \alpha$$

- Note: this covers the *region*, not the parameter τ itself.
 - Uses two-sided $z_{1-\alpha/2}$ (wider) vs. one-sided $z_{1-\alpha}$ for covering τ .